

1(a)(i)	concentration of, sodium chloride (solution) ;	1
1(a)(ii)	distilled water / deionised water / pure water (as diluent) ; correct volumes of stock solution and water (in Table 1.1) ;	2
1(b)(i)	any one from: 1 allows valid comparison (between onions in different sodium chloride concentrations) ; 2 allows comparison as, initial / starting, mass (of onions) is, not constant / AW ;	1
1(b)(ii)	any four from: 1 <i>idea that</i> the conclusion is only supported for onions left for hours ; ora 2 (ref. to intercept of x-axis / 0% change in mass / water potential of onions when left for) 48 hours is 3.1% sodium chloride ; 3 ref. to qualified experimental error ; 4 ref. to anomalous result(s) on graph ; 5 <i>idea that</i> line of best fit may give different value ; 6 ref. to intermediate concentrations (of sodium chloride) not tested (between 1% and 5% sodium chloride) ; 7 ref. to no, statistical analysis / standard error / 95% CI ; 8 <i>idea that</i> not all onion cells have the same water potential ;	4

Question	Answer	Marks
1(c)(i)	any six from: 1 use a minimum of five stated temperatures ; 2 suitable method to, maintain a constant temperature (for each replicate / experiment) ; 3 same / stated, variety / age / AW, of turnips ; 4 same / stated, dimensions / AW, of turnip block ; 5 ref. to apparatus for obtaining same size of turnip blocks ; 6 for each temperature, place turnip blocks in, a beaker / AW, of (fresh / new) distilled water ; 7 at each temperature, measure the initial mass and the final mass after a, set / stated, time ; 8 for each temperature, use at least three different blocks and calculate a mean ; 9 method to remove excess water from turnip block (before measuring final mass) ;	6
1(c)(ii)	1 <i>horizontal axis label:</i> temperature, °C and <i>vertical axis label:</i> rate of osmosis ; 2 appropriately shaped line ;	2

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1(c)(iii)	ref. to hazard and risk and precaution ; <table> <tr> <td>hazard</td><td>Risk</td><td>Precaution</td></tr> <tr> <td>knife / blade / AW</td><td>injury</td><td>cut away from the hand / cut down on to a, board / tile / AW</td></tr> <tr> <td>turnip</td><td>allergy irritant</td><td>wear gloves / PPE</td></tr> <tr> <td>hot water / water-bath / method of heating</td><td>scald / burn / injury</td><td>wear, (heat resistant) gloves / PPE use, tongs / AW</td></tr> </table>	hazard	Risk	Precaution	knife / blade / AW	injury	cut away from the hand / cut down on to a, board / tile / AW	turnip	allergy irritant	wear gloves / PPE	hot water / water-bath / method of heating	scald / burn / injury	wear, (heat resistant) gloves / PPE use, tongs / AW	1
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